

Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 2 · SLO 2 — Solved Worksheet

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Unit “ 1 : Session “ 2 : SLO “ 2 | Array of Structures

I. Fill in the Blanks

1. multiple records “ an array of structures stores many records of the same type.
2. `arrayName[size] “ struct StructName arr[100];`
3. `arr[index].member “` access a member via index and dot operator.
4. 5 Student structures “ `struct Student s[5];` declares an array of 5 Student records.
5. student/employee records “ e.g. roll numbers, names, salaries.
6. loop (for loop) “ read values into the array with a loop and `scanf`.
7. `emp[2].name “` the 3rd element's name field (index 2).
8. normal structure variable “ each element behaves like a regular struct variable.
9. `arr` (or `&arr[0]`) “ `struct StructName *ptr = arr;`
10. for loop “ loop with the dot operator prints each record's details.

II. Code Snippets

1. 100 students: `struct Student { int roll; char name[50]; }; struct Student s[100]; “ loop i=0..99 with scanf("%d %s", &s[i].roll, s[i].name).`
2. 5 employees: `struct Emp { int id; float salary; }; struct Emp e[5]; “ input loop then display loop printing e[i].id, e[i].salary.`
3. 10 students — 3 subjects: `struct Result { int marks[3]; }; struct Result r[10]; “ r[i].marks[j] holds subject j of student i.`
4. Shopping products: `struct Product { char name[50]; float price; }; struct Product p[50];`
5. Cricket team: `struct Player { char name[50]; int runs; }; struct Player team[11];`

III. Practice the Code

- `struct Car { char model[30]; int year; };`
- `struct Car c[3] = {{"Swift", 2020}, {"City", 2022}, {"Creta", 2023}};`
- `for (int i = 0; i < 3; i++) printf("%s (%d)\n", c[i].model, c[i].year);`